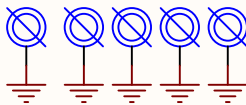
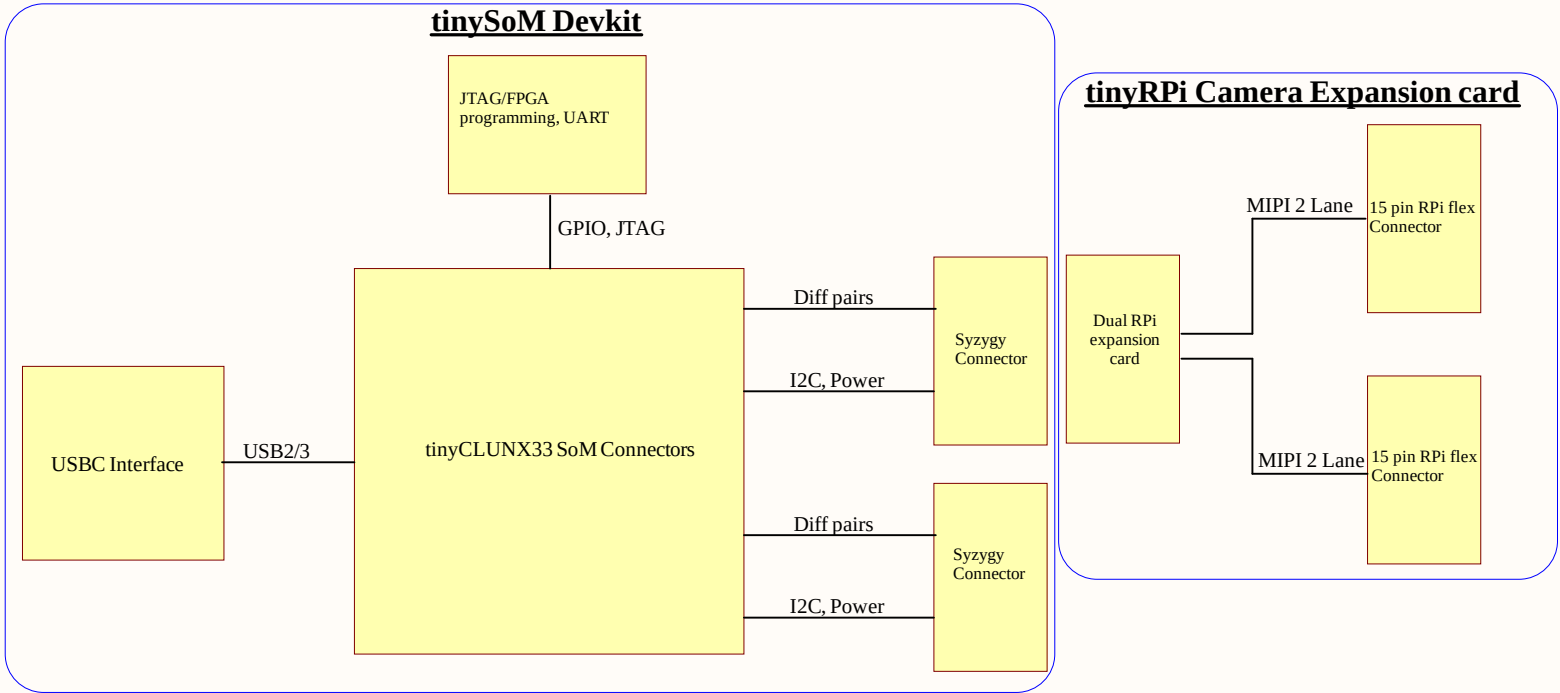


REV	Revision Notes	Designed by	Approved by	Date
1.0	Initial release	TinyVision	*	*
2.0	Rev2	TinyVision	*	*
3.0	Rev3	TinyVision	*	*



PCB
LOGO
WEEE

PCB
LOGO
ESD Susceptible



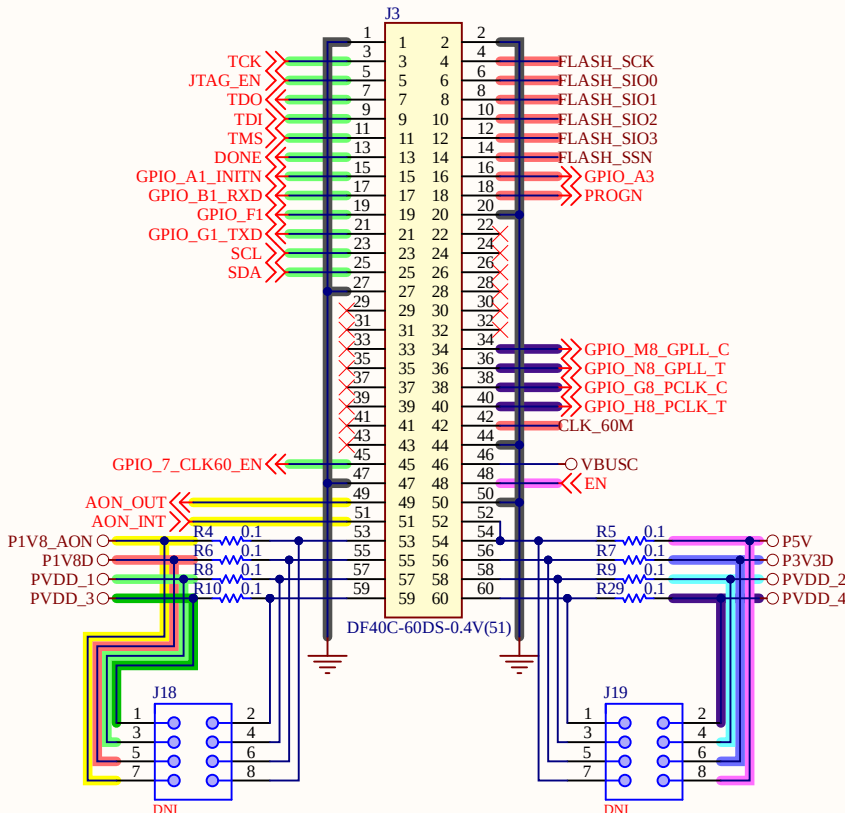
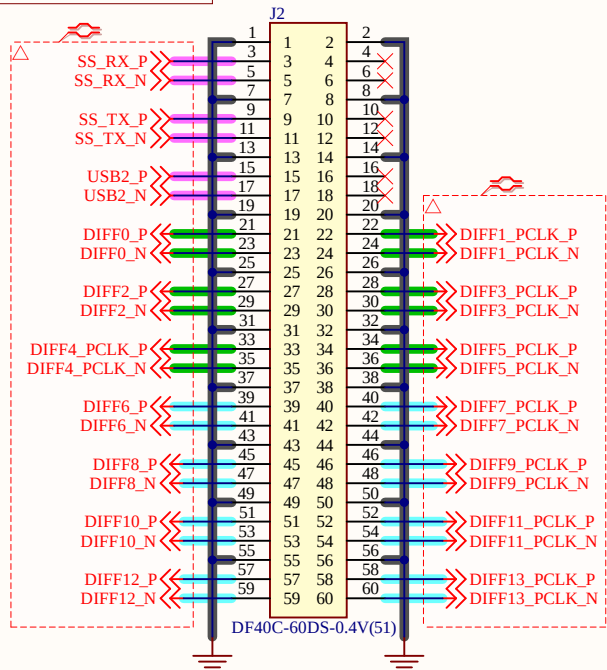
1 FID
1 FID
1 FID

Title <i>tinyNX33U Module BaseBoard</i>			
Size: B	Number: BlockDiagram	Revision: 4.0	
Date: 23-07-2026	Time: 16:47:35	Sheet 1 of 4	
File: Overview.SchDoc			

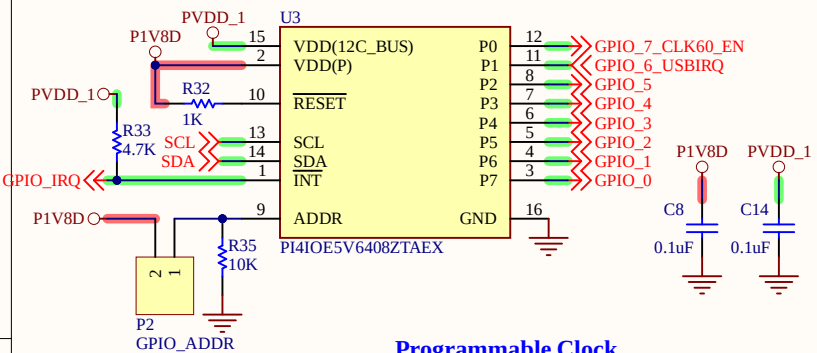
Color Legend:

3.3VIO
USB
1.8V
Bank 1 Voltage (1.2-3.3V)
Bank 2 Voltage (1.2-1.8V)
Bank 3 Voltage (1.2-1.8V)
Ground

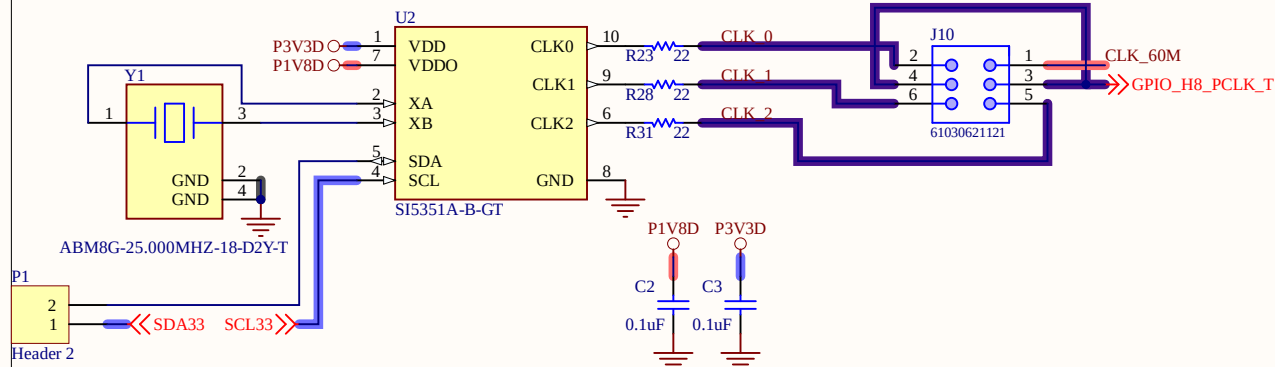
SoM Connectors



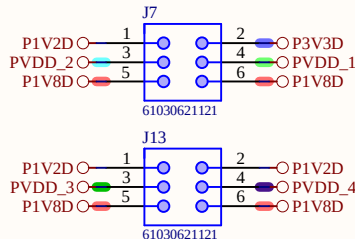
I2C GPIO Expander



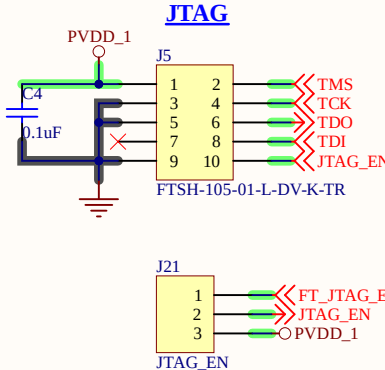
Programmable Clock



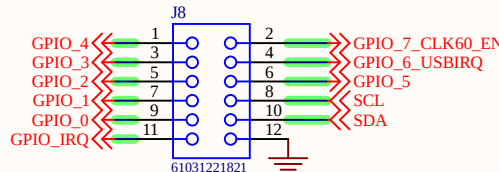
Bank Voltage Selection



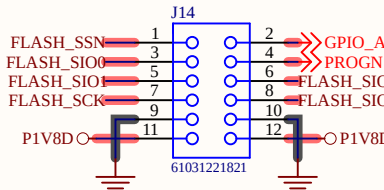
JTAG



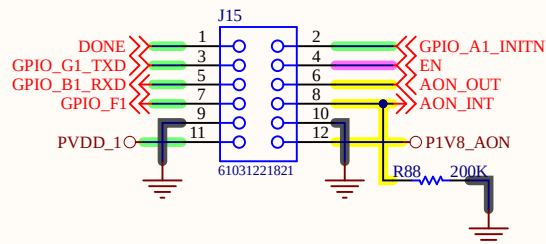
GPIO/I2C



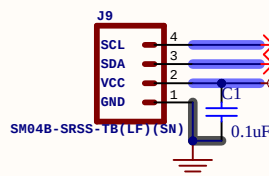
Flash IO



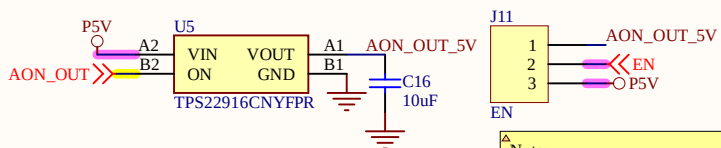
AON/GPIO



QWIIIC Connector



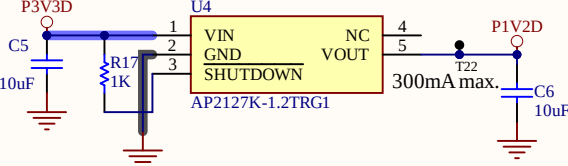
AON Circuit



Note:
This switch allows the EN line to go all the way to 5V allowing for more efficient PFM operation of the module switchers. Else, the switchers are limited to PWM (good for EMI but poor low load efficiency).

Note:
1-2: Power controlled by FPGA AON block
2-3: Power is always on

1.2V LDO



Note:
The 1.2V LDO is required as the bank voltage for proper MIPI operation.

Title **tinyNX33U Module BaseBoard**

Size: B

Number: BaseBoard_Connectors

Revision: 4.0

Date: 23-07-2026

Time: 16:47:35

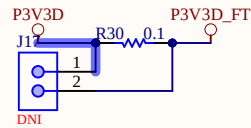
Sheet 2

of 4

File: Base_Module.SchDoc

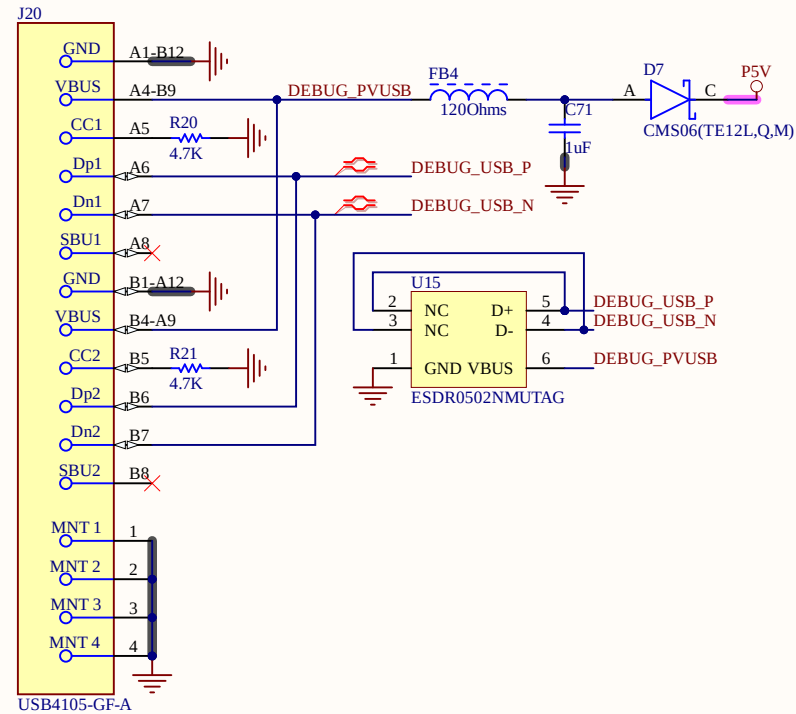
tinyvision.ai

FTDI Power Measurement/Disconnect

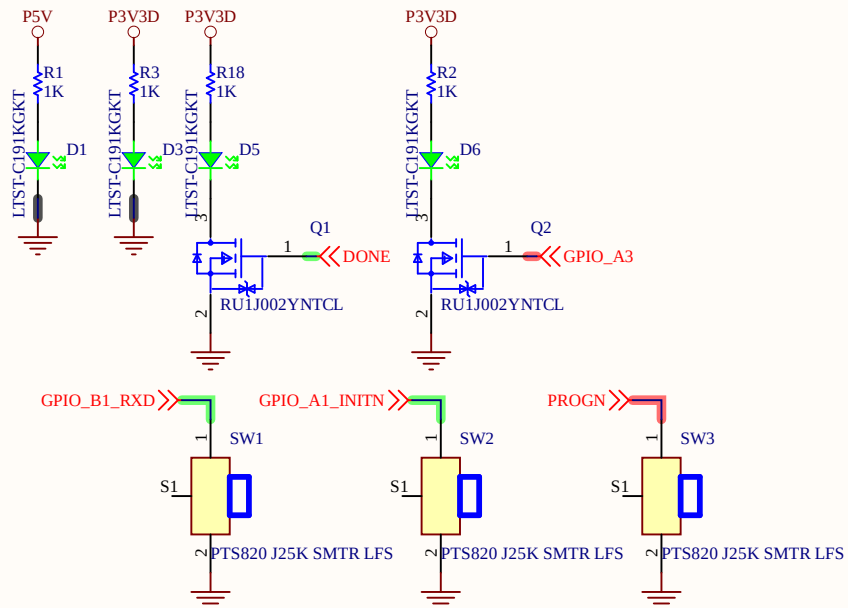


△ Note:
Can measure power into the FTDI or even disconnect the FTDI entirely by removing the current sense resistor.

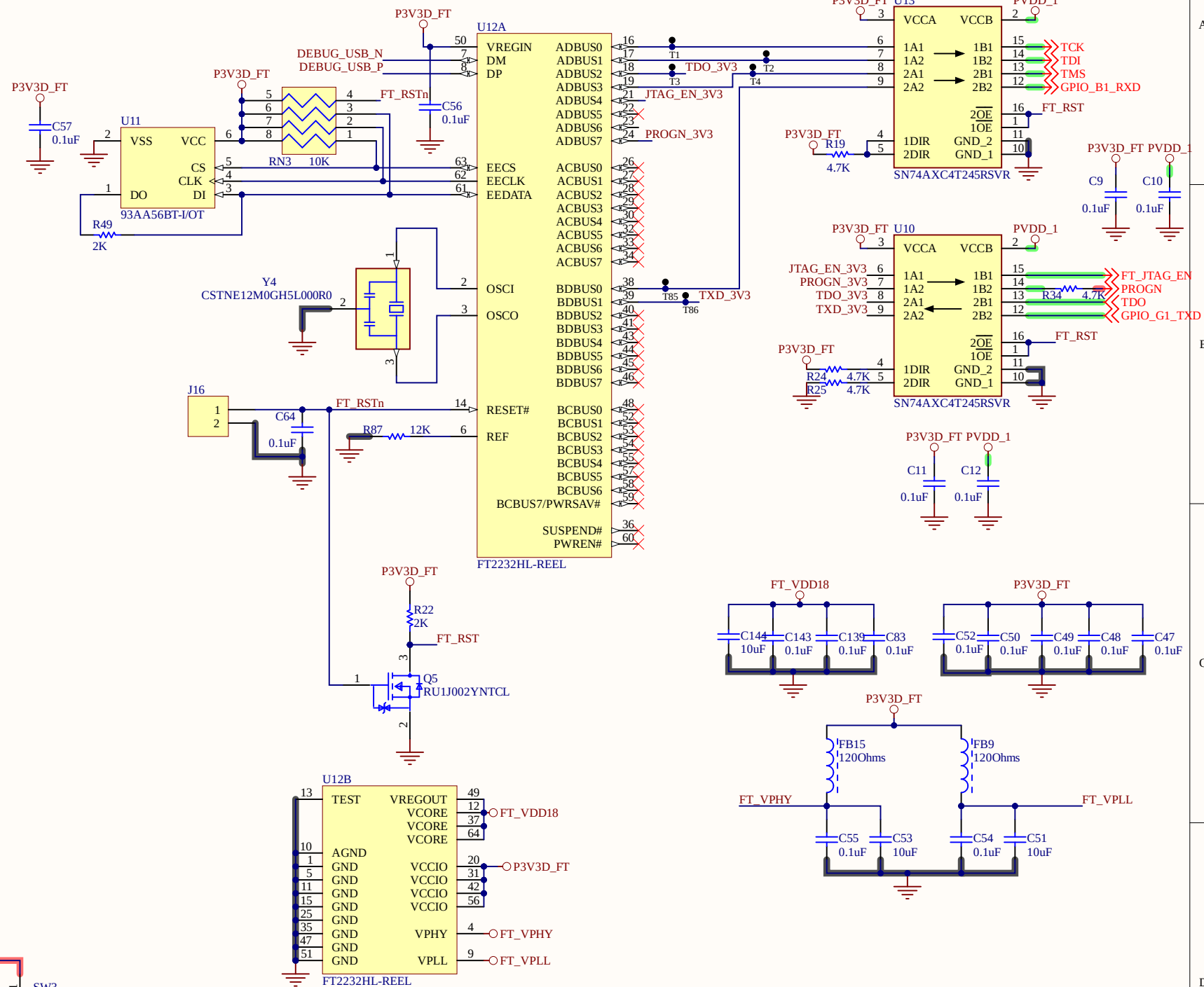
Debug/Programming USB



LED's/Switch



FPGA Programmer



Title <i>tinyNX33U Module BaseBoard</i>			
Size: B	Number: FPGA_Programmer	Revision: 4.0	
Date: 23-07-2026	Time: 16:47:35	Sheet 4 of 4	
File: FPGA_programmer.SchDoc			